TECHNICAL ENGLISH & WORKPLACE COMMUNICATION FOR TECHNICIANS

Electrical & HVAC Vocabulary, Manual Reading, Job Orders & Technical Interview English

MODULE 1: ESSENTIAL WORKSHOP & TECHNICAL VOCABULARY

A-Z ESSENTIAL TECHNICAL TERMS:
* Continuity: An uninterrupted electrical path for current flow without open-circuit breaks.
* Insulation Resistance: High resistance (Megaohms) preventing leakage current to chassis ground.
* Contactor: Electromagnetically operated heavy-duty switch for motor starting.
* Evacuation: The process of removing air, nitrogen, and non-condensable moisture using a vacuum pump.
* Flaring: Expanding copper pipe ends to a 45-degree cone for leak-proof brass flare nut mechanical joints.
* Micron Gauge: Precision digital vacuum instrument measuring absolute pressure down to single microns.

MODULE 2: READING MANUFACTURER TECHNICAL DATASHEETS

INTERPRETING NAMEPLATE SPECIFICATIONS:
* Rated Voltage / Frequency: e.g. 220-240V ~ 1 Phase 50Hz (Standard Sri Lankan & European supply).
* RLA (Rated Load Amps): Normal steady-state full load current during continuous compressor operation.
* LRA (Locked Rotor Amps): Extreme surge starting current (typically 5 to 6 times RLA for 0.5 seconds).
* Refrigerant Charge: e.g., R32 / 0.85 kg (Factory factory weight to be added strictly by electronic scale).

MODULE 3: PROFESSIONAL SERVICE REPORT & JOB CARD WRITING

SAMPLE TECHNICIAN INCIDENT / JOB REPORT:
"Customer Complaint: Indoor unit blowing warm air; error code E1 flashing on digital display.
Diagnosis: Inspected outdoor condenser unit. Found dual run capacitor bulging with zero capacitance reading.
Corrective Action: Replaced defective 35uF capacitor with genuine 450V component. Cleaned condenser coil.
Verification: Suction pressure stabilized at 125 PSIG (R410A). Delta-T across indoor coil measured at 11 deg C.
System operational and handed over to client in safe, optimal working condition."